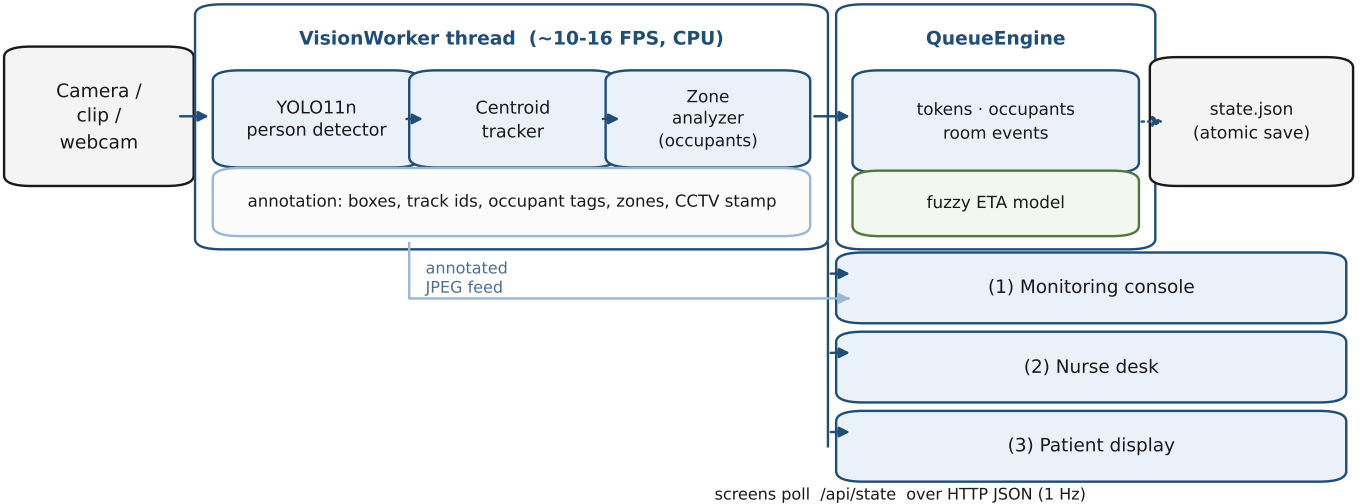
Camera / clip / webcam
VisionWorker thread (~10-16 FPS, CPU)
YOLO11n person detector
Centroid tracker
Zone analyzer (occupants)
annotation: boxes, track ids, occupant tags, zones, CCTV stamp
QueueEngine
tokens · occupants room events
fuzzy ETA model
state.json (atomic save)
annotated JPEG feed
(1) Monitoring console
(2) Nurse desk
(3) Patient display
screens poll /api/state over HTTP JSON (1 Hz)